

Institutional investors often fail to rebalance. While many pension funds and foundations resorted to panic selling and abandoned rebalancing during 2008 and 2009, CalPERS stands out in its failure. CalPERS's equity portfolio shrank from over \$100 billion in 2007 to \$38 billion in 2009.¹² CalPERS did the opposite of counter-cyclical rebalancing: it invested pro-cyclically and sold equities right at their lowest point—precisely when expected returns were highest. While part of CalPERS's problems in failing to rebalance stemmed from inadequate risk management, particularly of liquidity risk, CalPERS also failed to buy stocks when they were cheap because of structural misalignments between board members and the delegated fund manager. These are *agency problems*, and I discuss them in chapter 15. CalPERS did have a *statement of investment policy*, the institutional version of an individual investor's IPS, but this did not help CalPERS to rebalance during the financial crisis. CalPERS either didn't believe or optimally use its IPS.

In contrast to CalPERS, the Norwegian sovereign wealth fund rebalanced during 2008 and 2009. It bought equities at low prices from those investors like CalPERS who sold at the wrong time. Norway had its own version of Ulysses bound to the mast: the Ministry of Finance and parliament decided on a rebalancing rule, rather than having committees make rebalancing decisions. Adopting a rule, which was automatically implemented by the fund manager (Norges Bank Investment Management), ensured that the rebalancing decisions were not left to a committee whose members could be swayed by panic, fear, or hubris.

Rebalancing Bands

There are some technical considerations in implementing a rebalancing strategy. The theory presented has rebalancing occurring at regular time intervals: in Figure 4.5 and 4.6, rebalancing is done quarterly.¹³ But, in practice, if the equity portfolio weight is 61% at the end of a quarter, should the investor rebalance that small 1% back to a 60% target given transaction costs?

State-of-the-art rebalancing practices involve *contingent* rebalancing, rather than *calendar* rebalancing. Optimal rebalancing strategies trade off the utility losses of moving away from optimal weights versus the transaction costs from rebalancing. If the benefits of rebalancing outweigh the cost of doing so, then it is an optimal time to rebalance, and rebalancing becomes a contingent event.

Rebalancing bands are often used, set around optimal targets. The optimal rebalancing target may be 60% equities, for example, with bands set at 55% and 65%. A move outside the band triggers rebalancing. The bands are a function of transaction costs, liquidity, asset volatility, and minimum transaction sizes. When transaction costs are large or asset volatility is high, the bands are

¹² See "California Dreamin': The Mess at CalPERS," Columbia CaseWorks ID#120306.

¹³ The original Merton (1969, 1971) theory is presented in continuous time so rebalancing happens at every instant.